

Experiment - 14

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Aim - To study Amazon RDS, create a database instance on AWS RDS, understand security groups, and connect a cloud RDS database to a local database client such as pgAdmin.

Software used –

- AWS Management Console / AWS account
- Amazon RDS
- Amazon EC2 (for the MySQL-client approach shown in the guidelines)
- pgAdmin (for the PostgreSQL local-connection approach)
- Internet connection

Theory - Amazon RDS is a managed relational database service provided by AWS. The guidelines focus on scalability, performance, advanced security, cost-effectiveness, manageability, availability and durability.

The experiment contrasts databases running on EC2 with RDS. Running a database on EC2 can provide OS access, advanced DB option tuning, vendor-specific requirements and versions not provided by AWS, but it also introduces administration, backup/DR, monitoring and setup overhead.

Features –

Feature	Purpose
CloudWatch	Performance and health monitoring

Automated Backups	Disaster recovery
Manual Snapshots	User-created backups / long-term retention
Multi-AZ Deployment	High availability using a standby replica in another AZ
Read Replicas	Read-only copies for load balancing
Security (IAM, KMS, SSL)	Data protection, access control, compliance and safety
Performance Insights	SQL/load analysis and performance tuning
Storage Auto Scaling	Automatically grows storage as needed
Enhanced Monitoring	OS-level real-time metrics
Cross-Region Replicas	Replication for global availability
Parameter / Option Groups	Database configuration controls

Steps to create and configure the DB –

Go to AWS RDS Homepage:

Welcome to Aurora and RDS [Info](#)

Amazon Relational Database Service (Amazon RDS) and Amazon Aurora are fully managed database services that simplify setup, operation, and scaling of relational databases in the cloud.



Create with express configuration in seconds

Create and query an Aurora PostgreSQL serverless database with pre-configured settings to get started quickly.

[Create](#)



Create with full configuration

Create a database with custom configurations and enable features you want across a variety of engines including Aurora (compatible with MySQL & PostgreSQL), MySQL, PostgreSQL, MariaDB, Oracle, Microsoft SQL Server, and IBM Db2. Or use an existing database using restore from S3.

[Create](#)
[Restore from S3](#)

Select PostgreSQL provider:

Engine options

Engine type [Info](#)

☒ Aurora (MySQL Compatible)
 ☐ Aurora (PostgreSQL Compatible)
 ☐ MySQL
 ☒ PostgreSQL

☐ MariaDB
 ☐ Oracle
 ☐ Microsoft SQL Server
 ☐ IBM Db2

Choose a database creation method

☒ Full configuration
 You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
 Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates

Choose a sample template to meet your use case.

☐ Production
 Use defaults for high availability and fast, consistent performance.

☐ Dev/Test
 This instance is intended for development use outside of a production environment.

☒ Free tier
 Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

Set DB Instance Identifier:

Settings

Engine version [Info](#)

View the engine versions that support the following database features.

▼ Hide filters

☐ Show only versions that support the Multi-AZ DB cluster [Info](#)
 Create a Multi-AZ DB cluster with one primary DB Instance and two readable standby DB Instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.

Engine version

PostgreSQL 18.3-R2

☐ Enable RDS Extended Support [Info](#)
 Amazon RDS Extended Support is a paid offering. By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the RDS for PostgreSQL documentation.

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

KRG-DB

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

postgres

1 to 16 alphanumeric characters. The first character must be a letter.

Select Self-Managed and set Master Password

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure
 RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
 Create your own password or have RDS create a password that you manage.

☐ Auto generate password
 Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength [Average](#)

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

Select the required Instance type:

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ **Hide filters**

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

Instance type

db.t4g.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Storage

Storage type [Info](#)

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)
Baseline performance determined by volume size

Allocated storage [Info](#)

20

GiB

Allocated storage value must be 20 GiB to 6,144 GiB

► **Additional storage configuration**

Allow Public access:

Connectivity

[Info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

☒ **Don't connect to an EC2 compute resource**
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ **Connect to an EC2 compute resource**
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-072a3aacfd1569690)
3 Subnets, 3 Availability Zones

After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default

Public access [Info](#)

☒ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

Create custom VPC rules if required:

VPC security group (firewall) Info
 Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☐ Choose existing
 Choose existing VPC security groups

☒ Create new
 Create new VPC security group

New VPC security group name
 rds-practical-vpc

Availability Zone Info
 No preference

RDS Proxy
 RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.
☐ Create an RDS Proxy Info
 RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional Info
 Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.
 rds-ca-rsa2048-g1 (default)
 Expiry: May 25, 2061

If you don't select a certificate authority, RDS chooses one for you.

► Additional configuration

Wait for the DB to start

DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size	Recommendations	CPU
database-1	Available	Regional c...	Aurora Po...	SECOND	eu-north-1	0 instances		-
lrg-db	Creating	Instance	PostgreSQL	SECOND	-	db.t4g.micro		-

Open the InBound rules page:

Security Groups (1/1) Info Actions Export security groups to CSV Create security group

Find security groups by attribute or tag

rds-practical-vpc Clear filters

Name	Security group ID	Security group name	VPC ID	Description	Owner
-	sg-04eced45ea5c7529b	rds-practical-vpc	vpc-072a3aacfd1569690	Created by RDS management console	824648488987

sg-04eced45ea5c7529b - rds-practical-vpc

Details **Inbound rules** Outbound rules Sharing VPC associations Related resources - new Tags

Inbound rules (1) Manage tags Edit inbound rules

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sg-01a4e74773634d102	IPv4	PostgreSQL	TCP	5432	203.145.142.86/32	-

Add the host IP address

Inbound security group rules successfully modified on security group (sg-04eced45ea5c7529b | rds-practical-vpc)

Details

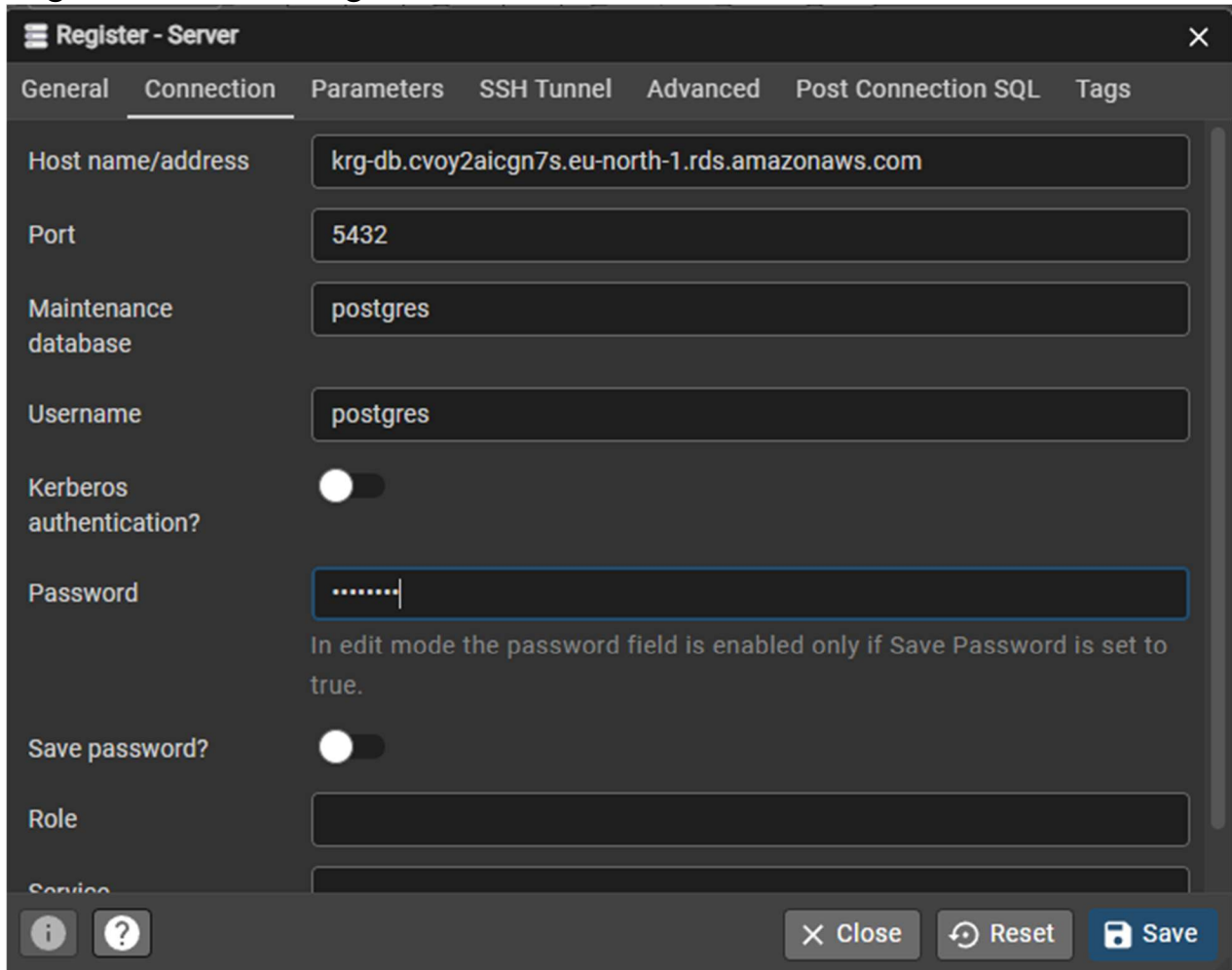
Security Groups (2) Info Actions Export security groups to CSV Create security group

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description	Owner
-	sg-052968687f09518ef	default	vpc-072a3aacfd1569690	default VPC security group	824648488987
-	sg-04eced45ea5c7529b	rds-practical-vpc	vpc-072a3aacfd1569690	Created by RDS management console	824648488987

Select a security group

Register the server in PgAdmin:

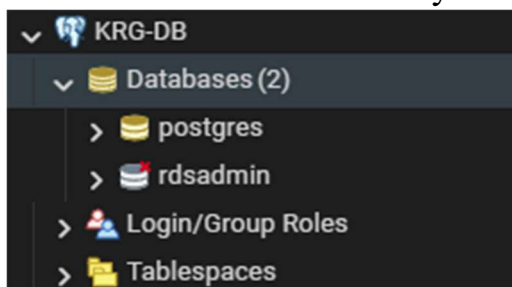


The image shows the 'Register - Server' dialog box in PgAdmin. The 'Connection' tab is selected. The fields are filled as follows:

Field	Value
Host name/address	kg-db.cvoy2aicgn7s.eu-north-1.rds.amazonaws.com
Port	5432
Maintenance database	postgres
Username	postgres
Kerberos authentication?	☐
Password	
Save password?	☐
Role	
Service	

At the bottom, there are buttons for 'Close', 'Reset', and 'Save'. A note below the password field states: 'In edit mode the password field is enabled only if Save Password is set to true.'

The DB is added successfully:



Verify the connection:

```
1 SELECT version();
```

Data Output

Messages

Notifications

≡+

▼

▼

SQL

	version text	
1	PostgreSQL 18.3 on aarch64-unknown-linux-gnu, compiled by aarch64-unknown-linux-gnu-gcc (GCC) 12.4.0, 6...	